

Semantic-Aware Multimodal Collaborative Learning for Unsupervised Visible-Infrared Person Re-Identification

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Abstract—Unsupervised visible-infrared person re-identification (VI-ReID) is challenging due to the significant modality gap between visible and infrared images. Most existing methods rely on one-hot clustering pseudo-labels as supervision signals, which often fail to capture the full semantic relationships among samples and are highly susceptible to noise. To address these limitations, we propose a Semantic-aware Multimodal Collaborative Learning (SAMCL) framework for unsupervised VI-ReID. Specifically, a Modality-aware Semantic Fusion (MSF) module is designed to bridge the inter-modality gap by integrating complementary semantic details from both visible and infrared modalities, generating enriched cross-modal supervision signals, for cross-modal collaborative learning. Meanwhile, we present a Dynamic Contrastive Learning (DCL) module to refine intra-modality feature learning by dynamically aligning samples with their neighboring centroids in the feature space, improving clustering reliability and intra-modality feature discrimination. By combining the two modules, SAMCL harnesses multimodal collaboration, minimizes dependence on noisy pseudo-labels, and provides a robust approach to unsupervised VI-ReID. Extensive experiments demonstrate the superiority of our proposed method. For instance, on the

SYSU-MM01 dataset, our model achieves a Rank-1 accuracy of 68.68% in the All Search setting, surpassing the state-of-the-art (SOTA) by 3.48%. On the RegDB dataset, it achieves a Rank-1 accuracy of 94.47% in the Visible-to-Infrared setting, outperforming the SOTA by 3.57%. On the LLCM dataset, it achieves a Rank-1 accuracy of 50.6% in the Visible-to-Infrared setting, outperforming the SOTA by 3.7%. The code is available at <https://github.com/luoshixi123/SAMCL>

Index Terms—Visible-infrared person re-identification, unsupervised learning, multimodal collaborative learning.

I. INTRODUCTION

PERSON re-identification aims to retrieve images of a specific individual from a set of images captured by multiple non-overlapping cameras [1], which is crucial for public security, helping to track criminals and locate missing persons. To support 24-hour surveillance, both visible and infrared cameras are used under different conditions [2], [3]. However, the significant modality gap between these two types of images creates challenges in learning effective features for cross-modal matching [4]. While some visible-infrared person re-identification methods have shown promise [5], they often rely on labor-intensive, well-annotated cross-modal data, which limits their practical applicability in real-world scenarios. Unsupervised visible-infrared person re-identification (USL-VI-ReID) enables cross-modal matching without the need for manual annotations [6], [7], [8], [9], making it a promising solution for continuous, real-time surveillance. Many existing methods rely on clustering algorithms like DBSCAN [10] to generate pseudo-labels, which are then used for training discriminative models. However, the generated *pseudo-labels are often imprecise and rigid*, leading to suboptimal performance.

Recent large-scale vision-language models like CLIP [11] excel at capturing high-level semantic features through powerful visual representations and image-text generation, making them highly effective for prompt learning in cross-modal tasks [12], [13], [14]. Unlike traditional unsupervised methods that rely on rigid pseudo-labels, CLIP generates semantically rich text descriptions, offering more flexible and detailed semantic supervision for the model optimization, as shown in Fig. 1. Inspired by this, CLIP-based person re-identification (ReID) methods have been proposed [14], [15], [16]. CSDN, for instance, adapts CLIP for person ReID by classifying

Received 9 January 2025; revised 18 July 2025, 14 January 2026, and 1 April 2026; accepted 7 April 2026. Date of publication 23 April 2026; date of current version 14 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62573189 and Grant 62425305, in part by the Science and Technology Innovation Program of Hunan Province under Grant 2024RC3151 and Grant 2023RC1048, and in part by the Natural Science Foundation of Hunan Province under Grant 2024JJ3013 and Grant 2026JJ50511. The associate editor coordinating the review of this article and approving it for publication was Dr. Zhengming Ding. (Corresponding authors: Min Liu; Gautam Srivastava; Shuai Liu.)

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Digital Object Identifier 10.1109/TIP.2026.3684860

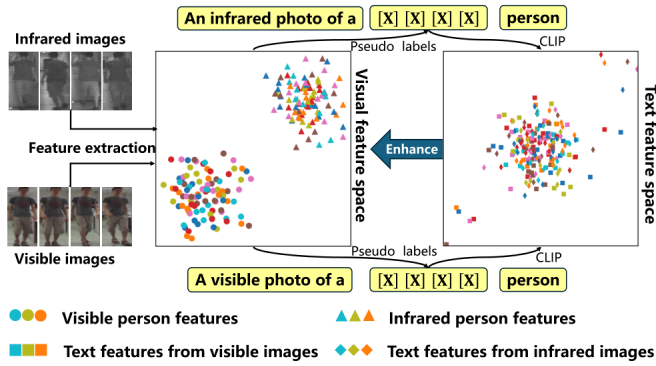


Fig. 1. Illustration of the modality gap in the VI-ReID task. Distinct shapes represent features from different modalities, with visible and infrared features widely separated in the visual feature space due to differing imaging principles. Conversely, textual descriptions in the text feature space generated by CLIP show smaller inter-modal differences, motivating the integration of textual semantics to enhance visual feature representations, improve recognition accuracy, and increase robustness. Best viewed in color.

images into ID-specific prompts [17], introducing a prompt learning approach for unsupervised visible-infrared ReID [18]. It creates learnable, cluster-aware prompts and generates text descriptions for training. However, due to *the modality differences between visible and infrared features*, CLIP can only generate text descriptions from single-modal prompts. Its pretraining objective is primarily designed for global image-text alignment, and the visible-domain prior it inherits can introduce semantic bias when transferred to infrared-visible identity matching tasks.

To address the above issues, in this paper, we propose a Semantic-Aware Multimodal Collaborative Learning framework for unsupervised visible-infrared person re-identification, which tackles two core challenges: modality gap and noisy pseudo-labels. Specifically, to overcome the modality gap in unsupervised VI-ReID, we propose a Modality-aware Semantic Fusion (MSF) module. Inspired by prior works, we adopt a prompt learning strategy to generate corresponding textual semantic information [17], [18] for the clustering pseudo-labels of both visual and infrared modality data, separately. MSF then integrate complementary textual features from the two modalities. Specifically, visible textual semantics are used as queries and infrared textual semantics as keys/values to construct fused semantic representations, which are then used to refine the modality-specific textual features. Through this process, cross-modal textual supervision signals are reconstructed with richer semantic information, offering strong and reliable guidance for cross-modal collaborative training. Also, this design helps compensate for potential biases or under-representation of infrared-specific semantics in CLIP. Moreover, to mitigate the impact of noisy pseudo-labels, we design a Dynamic Contrastive Learning module that enhances feature consistency within each modality by aligning each sample with its k -nearest neighbors—pulling similar features closer while pushing dissimilar ones apart—thereby stabilizing learning and reducing clustering noise. Together, these strategies reduce reliance on unreliable clustering pseudo-labels, offering a more robust solution for unsupervised VI-ReID.

To summarize, the contributions of our work are as follows.

- We propose a Modality-aware Semantic Fusion module that captures higher-order semantic information as supervision signals by integrating textual features from both visible and infrared language descriptions for the unsupervised VI-ReID task.
- We present a Dynamic Contrastive Learning module that enhances intra-modality learning by dynamically measuring distances between samples and their neighboring centroids, improving clustering reliability and intra-modality feature discrimination.
- Our method achieves state-of-the-art performance on three widely-used VI-ReID datasets such as SYSU-MM01, RegDB and LLCM.

II. RELATED WORK

A. Visible-Infrared Person Re-Identification

VI-ReID is a cross-modality task in the field of person re-identification, aiming to match images of the same person captured under visible and infrared conditions. In addition to the challenges posed by traditional person re-identification tasks, such as variations in lighting, camera angles, and occlusion, VI-ReID is further complicated by the significant modality differences between visible and infrared images [19], [20], [21], [22], [23], [24]. Several techniques have been proposed to address these difficulties. The MSO method [19] integrates multi-feature space optimization, multi-level fusion, and adversarial modality alignment within an end-to-end framework. PartMix [21] introduces a partitioned mixing regularization to facilitate cross-modal body part feature discovery and modality alignment. DEEN [25] proposes a Diverse Embedding Expansion Network that expands the embedding space to obtain diverse modality-specific representations, along with a low-light cross-modality regularization for robust feature modeling under extreme lighting conditions. However, these supervised methods require large annotated datasets for training, which incurs high costs and poses practical challenges. Our method adopts an unsupervised approach, avoiding the expensive costs of data annotation, making it more feasible for real-world applications.

B. Unsupervised Visible-Infrared Person Re-Identification

The advantage of unsupervised visible-infrared person re-identification lies in its ability to learn both modality-specific information and modality-invariant information without requiring cross-modal annotations. As a result, extensive research and exploration have been conducted. First, reliable pseudo-labels are obtained through clustering methods [26], followed by cross-modal paired instance selection to align the two modalities. For instance, H2H [27] primarily focuses on homogeneous learning to acquire modality-specific features, and then conducts heterogeneous learning to extract shared modality knowledge. OTLA [28], on the other hand, employs optimal transport methods to establish connections between different modalities. Recent approaches, such as ADAC [29], have advanced unsupervised VI-ReID by leveraging random channel enhancement and dual contrastive learning techniques.

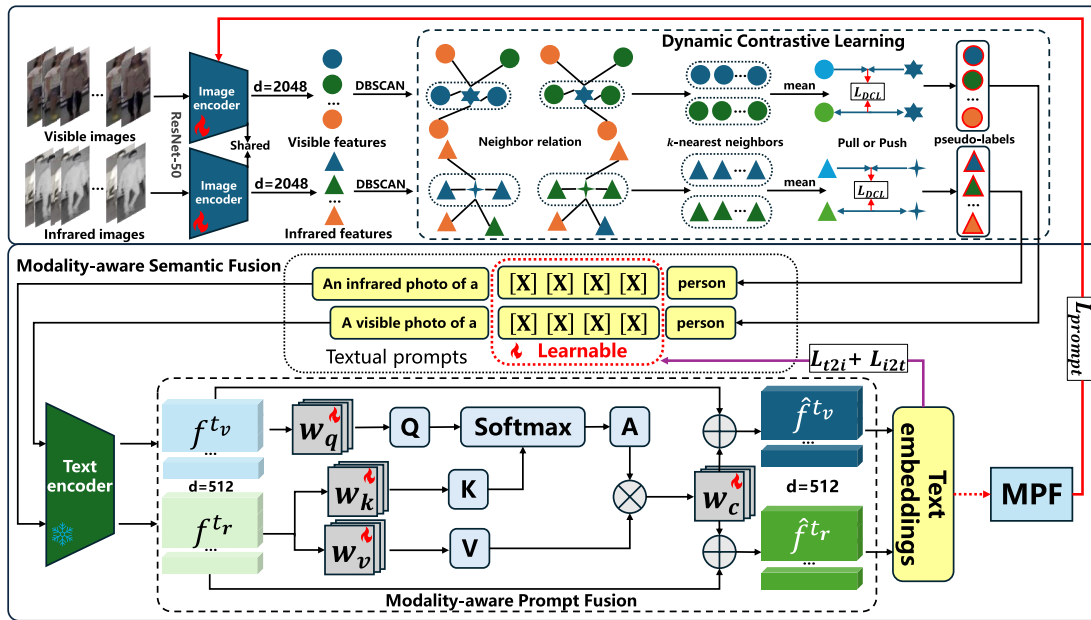


Fig. 2. Illustration of the semantic-aware multimodal collaborative learning (SAMCL) framework for unsupervised VI-ReID. It features two key modules: Modality-aware semantic fusion, which uses CLIP-generated unimodal textual features to produce semantic supervision signals for cross-modal representation, and dynamic contrastive learning, which improves clustering and feature discrimination through sample-to-neighbor dynamics. MPF refers to the Modality-aware prompt fusion module. The red arrows indicate the loss computation during ReID model optimization, while the purple arrows represent the loss calculation during the prompt optimization stage, flames denote parameter updates, and ice symbols represent frozen parameters.

PGM [30] has developed a progressive graph matching method to systematically extract cross-modal correspondences and define the correspondence mining process as graph matching. OTLA [28] and DOTLA [31] adopt an optimal transport strategy to assign pseudo-labels from one modality to another at the instance level. However, these methods only use one-hot vectors as supervision signals, without considering the potential of textual information as a supervisory signal, which limits their effectiveness. We introduce CLIP, which generates strong supervision signals using textual information, significantly improving the performance of unsupervised models.

C. CLIP-Based Person Re-Identification

Large-scale pre-trained vision-language models like CLIP have shown exceptional transfer capabilities for downstream tasks, leveraging textual image descriptions instead of relying solely on discrete labels [32]. Li et al. are the first to employ the CLIP-ReID model [16], generating language descriptions related to person identities through learnable prompts, with the visual encoder extracting semantically rich image features under the supervision of natural language. Yan et al. apply the CLIP-ReID model to cross-modality person re-identification, proposing the CSDN framework [14], which uses CLIP prompts to generate textual descriptions for each modality and then fuses these descriptions using a self-attention module before conducting contrastive learning between the text-image features and the images themselves. Chen et al. introduce CLIP into unsupervised cross-modality person re-identification, proposing the CCLNet framework [18], which generates textual features for each modality through a cluster-aware prompt learning method, achieving promising results. However, these CLIP-based approaches primarily rely on unimodal textual descriptions and overlook two critical

challenges inherent to unsupervised VI-ReID: the misalignment between CLIP's coarse semantic supervision and the fine-grained identity discrimination required for ReID, and the instability of intra-modality feature structures under noisy clustering. To address these limitations, we introduce two coordinated components: a Modality-aware Semantic Fusion module, which adapts and fuses modality-specific textual semantics to provide effective cross-modal supervision, and a Dynamic Contrastive Learning module, which stabilizes intra-modality neighborhood structures despite the presence of noisy pseudo labels.

III. PROPOSED METHOD

This paper proposes a Semantic-aware Multimodal Collaborative Learning framework for the unsupervised visible-infrared person re-identification task. As illustrated in Fig. 2, the framework consists of two key modules: the Modality-aware Semantic Fusion module (MSF) and the Dynamic Contrastive Learning (DCL) module. In MSF, the text encoder of the pre-trained CLIP model generates unimodal textual features (visible and infrared modality) for pedestrian images, which are fused to produce advanced, modality-specific semantic information as supervision signals, enhancing cross-modal representation. DCL dynamically adjusts the learning process by leveraging the differences between each sample and its neighbors within the same modality, improving clustering reliability and intra-modality feature discrimination.

A. Modality-Aware Semantic Fusion

In recent years, large-scale pre-trained vision-language models like CLIP have shown significant advantages in generating textual descriptions for images. These descriptions offer richer semantic insights than traditional fixed labels and

are cost-effective to obtain without additional supervision or domain knowledge. This subsection argues that such textual descriptions provide superior semantic supervision compared to pseudo-labels generated by common clustering methods in the unsupervised VI-ReID task.

Currently, most unsupervised Visible-Infrared Re-identification methods typically employ clustering algorithms to generate pseudo-labels which are then used to train models with discriminative capabilities. Within this framework, we first extract visible features $\{f_i^v\}_{i=1}^{N_v}$ and infrared features $\{f_i^r\}_{i=1}^{N_r}$ from all visible images X^v and infrared images X^r using an image encoder of a pre-trained CLIP model. Thereafter, DBSCAN [10] is employed to assign pseudo-labels to all images, denoted as $Y^v = \{y_1^v \dots y_m^v\}$ and $Y^r = \{y_1^r \dots y_n^r\}$, respectively. Note that y_i^v denotes the i -th clustering label for visible images, similarly y_i^r for infrared images. However, clustering pseudo-labels are often inaccurate and rigid, limiting progress in unsupervised VI-ReID. To address this, we propose to leverage the capabilities of the CLIP model to generate textual information to complement the limitations of rigid pseudo-labels, providing valuable semantic insights.

1) *Modality-Aware Prompt Generation*: For each cluster y_i^v or y_i^r generated by DBSCAN, we define the textual prompts as “A visible photo of a $[X_i]_1[X_i]_2 \dots [X_i]_M$ person.” and “An infrared photo of a $[X_i]_1[X_i]_2 \dots [X_i]_M$ person.”. Here, $[X_i]$ is a randomly initialized learnable token for the i -th cluster, and M represents the number of learnable textual tokens.

Then, we feed the textual prompts and images into the text encoder and image encoder of a pre-trained CLIP model, respectively. This enables us to obtain unique textual class semantic representations for each cluster, denoted as $F^{t_v} = \{f_1^{t_v} \dots f_m^{t_v}\}$ and $F^{t_r} = \{f_1^{t_r} \dots f_n^{t_r}\}$, herein, F^{t_v} represents the visible modality, and F^{t_r} the infrared modality.

2) *Modality-Aware Prompt Fusion*: Considering the large modality gap between visible and infrared images, we introduce the Modality-aware Prompt Fusion module that integrates the obtained textual class semantic information F^{t_v} and F^{t_r} into the comprehensive class semantic prompts. Specifically, F^{t_v} serves as the query, and F^{t_r} functions as both key and value. The textual semantic fusion operates as follows:

$$Q = W_q \cdot F^{t_v}, K = W_k \cdot F^{t_r}, V = W_v \cdot F^{t_r} \quad (1)$$

$$F^{t_x} = W_c \cdot \left(\text{Softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V \right) \quad (2)$$

where d is the dimensionality of the textual features. W_q , W_k , W_v , and W_c correspond to four fully connected layers. F^{t_x} represents the fused textual semantic information, which contains the complementary semantic details of visible and infrared modalities. Building on the fusion attention process, modality-based weighting refines the visible and infrared features into modality-aware semantic representations. This results in enhanced textual semantic features specific to each modality, providing a strong supervisory signal $\hat{F}^{t_v} = \{\hat{f}_1^{t_v} \dots \hat{f}_m^{t_v}\}$ and $\hat{F}^{t_r} = \{\hat{f}_1^{t_r} \dots \hat{f}_n^{t_r}\}$ for multimodal collaborative learning.

$$\hat{F}^{t_v} = F^{t_v} + \lambda F^{t_x} \quad (3)$$

$$\hat{F}^{t_r} = F^{t_r} + \lambda F^{t_x} \quad (4)$$

where λ is a hyperparameter.

3) *Prompt Optimization*: Note that in this training stage, we freeze the image encoder and the text encoder, optimizing only the textual tokens $[X_i]$ and the attention layers W_q , W_k , W_v , and W_c . To ensure a one-to-one correspondence between the textual features $(\hat{f}_i^v, \hat{f}_i^r)$ and the visual features (f_i^v, f_i^r) , we initially adopt a visual-to-text loss L_{i2t} and a text-to-visual loss L_{t2i} in CLIP.

$$L_{i2t} = -\frac{1}{N_b/2} \sum_{i=1}^{N_b/2} \log \frac{\exp(\langle f_i^v, \hat{f}_i^{t_v} \rangle / \tau)}{\sum_{j=1}^{N_b/2} \exp(\langle f_i^v, \hat{f}_j^{t_v} \rangle / \tau)} - \frac{1}{N_b/2} \sum_{i=1}^{N_b/2} \log \frac{\exp(\langle f_i^r, \hat{f}_i^{t_r} \rangle / \tau)}{\sum_{j=1}^{N_b/2} \exp(\langle f_i^r, \hat{f}_j^{t_r} \rangle / \tau)} \quad (5)$$

$$L_{t2i} = -\frac{1}{N_b/2} \sum_{i=1}^{N_b/2} \log \frac{\exp(\langle \hat{f}_i^{t_v}, f_i^v \rangle / \tau)}{\sum_{j=1}^{N_b/2} \exp(\langle \hat{f}_i^{t_v}, f_j^v \rangle / \tau)} - \frac{1}{N_b/2} \sum_{i=1}^{N_b/2} \log \frac{\exp(\langle \hat{f}_i^{t_r}, f_i^r \rangle / \tau)}{\sum_{j=1}^{N_b/2} \exp(\langle \hat{f}_i^{t_r}, f_j^r \rangle / \tau)} \quad (6)$$

where f_i^v and f_i^r represent the features of visible and infrared images, respectively. $\hat{f}_i^{t_v}$ and $\hat{f}_i^{t_r}$ denote the corresponding fused textual features. N_b is the batch size, $\langle, \rangle$ represents the cosine similarity between two vectors, and τ is determined by the temperature coefficient. We set $\tau = 1.0$ in the experiments.

Considering different images within a batch may belong to the same individual, we adjust the text-to-visual loss:

$$L_{t2i} = -\frac{1}{N_b/2} \sum_{i=1}^{N_b/2} \frac{1}{|S(y_i)|} \sum_{s \in S(y_i)} \log \frac{\exp(\langle \hat{f}_{y_i}^{t_v}, f_s^v \rangle / \tau)}{\sum_{j=1}^{N_b/2} \exp(\langle \hat{f}_{y_i}^{t_v}, f_j^v \rangle / \tau)} - \frac{1}{N_b/2} \sum_{i=1}^{N_b/2} \frac{1}{|S(y_i)|} \sum_{s \in S(y_i)} \log \frac{\exp(\langle \hat{f}_{y_i}^{t_r}, f_s^r \rangle / \tau)}{\sum_{j=1}^{N_b/2} \exp(\langle \hat{f}_{y_i}^{t_r}, f_j^r \rangle / \tau)} \quad (7)$$

where $\hat{f}_{y_i}^{t_v}$ and $\hat{f}_{y_i}^{t_r}$ represent the fused textual features with identity y_i ; $S(y_i)$ denotes a set composed of sample indices with identity y_i ; $|S(y_i)|$ represents the cardinality of $S(y_i)$. Therefore, we optimize the learnable prompt jointly with the image-to-text loss L_{i2t} and modified text-to-image loss L_{t2i} .

B. Dynamic Contrastive Learning

Although CLIP model generates robust supervisory signals by providing rich textual semantic information, the training process inevitably includes noisy samples due to they fail to ensure that similar inputs from the same modality stay close by. Therefore, we design a Dynamic Contrastive Learning module (DCL) to enhance feature consistency within each modality (visual or infrared modality) by aligning each sample with its k -nearest neighbors, pulling similar features closer and pushing dissimilar ones apart to stabilize learning and reduce clustering noise. Specifically, this module identifies

Algorithm 1 Semantic-Aware Multimodal Collaborative Learning

Input: Visible images X^v and infrared images X^r . Frozen visual encoder $\mathcal{I}$ and text encoder $\mathcal{T}$ of a pre-trained CLIP model. Initialized ReID model $f(\cdot)$. The number of iterations of prompt optimization N_1 and ReID model optimization N_2 .

Output: Well-trained ReID model.

- 1: Modality-aware semantic fusion
- 2: **for** $iter = 1$ to N_1 **do**
- 3: Extract visible and infrared features:
- 4: $\mathcal{I} : X^v \rightarrow \{f_i^v\}_{i=1}^{N_v}$,
- 5: $\mathcal{I} : X^r \rightarrow \{f_i^r\}_{i=1}^{N_r}$.
- 6: Generate clustering pseudo-labels using DBSCAN:
- 7: $\{f_i^v\}_{i=1}^{N_v} \rightarrow \{y_1^v, \dots, y_m^v\}$,
- 8: $\{f_i^r\}_{i=1}^{N_r} \rightarrow \{y_1^r, \dots, y_n^r\}$.
- 9: Generate modality-aware prompts with $\mathcal{T}$:
- 10: $F^{tv} = \{f_1^{tv}, \dots, f_m^{tv}\}$,
- 11: $F^{tr} = \{f_1^{tr}, \dots, f_n^{tr}\}$.
- 12: Fuse modality-aware semantic information:
- 13: $\hat{F}^{tv}, \hat{F}^{tr}$ based on Eqs. (1)-(4).
- 14: Optimize prompts using Eqs. (5) and (7).
- 15: **end for**
- 16: Semantic-driven ReID model optimization
- 17: **for** $iter = 1$ to N_2 **do**
- 18: Extract visible and infrared features:
- 19: $f(\cdot) : X^v \rightarrow \{f_i^v\}_{i=1}^{N_v}$,
- 20: $f(\cdot) : X^r \rightarrow \{f_i^r\}_{i=1}^{N_r}$.
- 21: Calculate target embedding with Eq. (8).
- 22: Distance regularization using Eq. (9).
- 23: Calculate contrastive loss L_{DCL} with Eq. (12).
- 24: Calculate semantic-guided loss L_{prompt} with Eq. (15).
- 25: Optimize the ReID model with Eq. (16).
- 26: **end for**

the k -nearest neighbors of a given sample to determine a target embedding and compute its similarity to the original embedding. Focusing on visible samples, the top k nearest neighbors $NN_k(f_i^v)$ of feature f_i^v are calculated based on the Jaccard similarity. The target embedding $\tilde{f}_i^v$ is then obtained by calculating the mean of the selected top k nearest neighbors:

$$\tilde{f}_i^v = \frac{1}{k} \sum_{h \in NN_k(f_i^v)} f_h^v \quad (8)$$

To better control the collaborative consistency, a Gaussian kernel function w_{ij}^v is introduced to regulate the distance between the target embedding $\tilde{f}_i^v$ and the source features f_i^v as below:

$$w_{ij}^v = \exp\left(-\frac{(s_{ij}^v)^2}{\sigma}\right) \quad (9)$$

where s_{ij}^v is the Euclidean distance between $\tilde{f}_i^v$ and f_i^v . The parameter σ controls the bandwidth of the Gaussian kernel. We set $\sigma = 1.0$ in the experiments.

On top of that, we design a dynamic learning loss. If the source and target features share substantial collaborative information, their distance should be minimized to ensure that the source sample aligns more closely with its neighboring samples, thereby enhancing collaborative consistency. To achieve this, an attractive term is introduced to reduce the distance between similar sample pairs.

$$L_{pull}^v = \frac{1}{N_b} \sum_{i=1}^{N_b} \sum_{j=1, j \neq i}^{N_b} w_{ij}^v \cdot (s_{ij}^v)^2 \quad (10)$$

Conversely, when the source and target features share minimal collaborative information, it is beneficial for them to maintain a greater distance in the feature space. To facilitate

this, a repulsion term is added to increase the distance between dissimilar sample pairs, specifically applying the repulsive force to negative pairs.

$$L_{push}^v = \frac{1}{N_b} \sum_{i=1}^{N_b} \sum_{j=1, j \neq i}^{N_b} (1 - w_{ij}^v) \cdot (\max(\delta - s_{ij}^v, 0))^2 \quad (11)$$

where N_b represents the batch size. δ represents a predefined threshold, indicating the minimum desired distance to be maintained between dissimilar embeddings. We set $\delta = 1.0$ in the experiments.

Therefore, the dynamic contrastive loss for visible images is formulated as $L_{DCL}^v = L_{pull}^v + L_{push}^v$, similarly $L_{DCL}^r = L_{pull}^r + L_{push}^r$ for infrared images. The dynamic contrastive loss is as below:

$$L_{DCL} = L_{DCL}^v + L_{DCL}^r \quad (12)$$

C. Semantic-Driven ReID Model Optimization

Our primary objective is to refine visual representations for precise identity matching in VI-ReID tasks. To achieve this, we not only use clustering-based pseudo-labels as supervision signals for model optimization, incorporating identity loss (L_{id}) and triplet loss (L_{tri}), but also leverage class textual semantic features $\hat{F}_{t_v}$ and $\hat{F}_{t_r}$ as additional supervision signals. To enhance the effectiveness of these textual features for each modality, we adopt the modality-aware prompt fusion (MPF) strategy outlined in Section A.(2) to refine the textual features into $\tilde{F}^{tv} = \{\tilde{f}_1^{tv}, \dots, \tilde{f}_m^{tv}\}$ for the visual modality and $\tilde{F}^{tr} = \{\tilde{f}_1^{tr}, \dots, \tilde{f}_n^{tr}\}$ for the infrared modality. The fused textual semantic information is then used to align cross-modal features, facilitating accurate identity matching. Additionally, we employ this fused textual information to compute an image-to-text contrastive loss, further improving the alignment between modalities and enhancing model performance.

$$L_{i2\ tce}^v = -\frac{1}{N_b} \sum_{i=1}^{N_b} q_i \log \frac{\exp \langle f_i^v, \tilde{f}_{y_i}^{tv} \rangle}{\sum_{y_a=1}^m \exp \langle f_i^v, \tilde{f}_{y_a}^{tv} \rangle} \quad (13)$$

$$L_{i2\ tce}^r = -\frac{1}{N_b} \sum_{i=1}^{N_b} q_i \log \frac{\exp \langle f_i^r, \tilde{f}_{y_i}^{tr} \rangle}{\sum_{y_a=1}^n \exp \langle f_i^r, \tilde{f}_{y_a}^{tr} \rangle} \quad (14)$$

Here, q_i represents the one-hot vector of the identity label y_i . This loss function is particularly designed to train the visual encoder, allowing the model to achieve optimal recognition performance through textual supervision signals. Therefore, the textual semantic-guided loss can be formulated as below:

$$L_{prompt} = L_{i2\ tce}^v + L_{i2\ tce}^r \quad (15)$$

The total training loss for the VI-ReID model optimization can be formulated as follows:

$$L_{total} = L_{id} + L_{tri} + \alpha_1 L_{DCL} + \alpha_2 L_{prompt} \quad (16)$$

IV. EXPERIMENTS

A. Dataset and Evaluation Metrics

1) *Dataset:* We evaluate the proposed Semantic-Aware Multimodal Collaborative Learning framework on three

widely-used visible-infrared person re-identification datasets, SYSU-MM01 [50], RegDB [51] and LLCM [25].

The **SYSU-MM01** dataset is widely recognized as one of the most demanding benchmarks in visible-infrared person re-identification. It consists of images captured from six cameras, including four visible and two infrared, across a variety of real-world settings. The dataset is divided into a training set with 395 identities and a testing set with 96 identities, containing a total of 30,071 visible images and 22,258 infrared images.

The **RegDB** dataset is a small-scale benchmark for visible-infrared person re-identification, comprising 8,240 images of 412 identities. Each identity is represented by 10 visible and 10 infrared images, all captured by a single camera per modality. Following the standard evaluation protocol, 2,060 visible and 2,060 infrared images from 206 identities are randomly selected for training, while the remaining images are used for testing.

The **LLCM** dataset is a challenging low-light benchmark for visible-infrared person re-identification, collected by a camera network consisting of 9 visible cameras and 9 infrared cameras. It comprises 46,767 person images from 1,064 identities. Following the official protocol, LLCM is split into a training set and a testing set with an approximate ratio of 2:1. Specifically, the training set contains 30,921 images from 713 identities, while the testing set includes 13,909 images from 351 identities.

2) *Evaluation Metrics*: To thoroughly evaluate the proposed method, we employ several metrics, including Cumulative Matching Characteristics (CMC), mean Average Precision (mAP), and mean Inverse Negative Penalty (mINP) [5]. For the SYSU-MM01 dataset, we conduct evaluations in two modes: All-Search and Indoor-Search. For the RegDB and LLCM dataset, performance is assessed in two retrieval modes: Visible-to-Infrared (Visible2Infrared) and Infrared-to-Visible (Infrared2Visible). To ensure fairness and stability, we average the experimental results over 10 random splits of the training and testing sets.

B. Implementation Details

We use a modified ResNet-50 backbone pretrained on CLIP [11] as our feature extraction network. The input image size is set to 288×144 , and we apply data augmentation techniques such as random flipping, random gray scaling, and random erasing to enhance the diversity of the training data [52], [53]. The training process is divided into two distinct optimization stages. Stage 1: Modality-aware Semantic Fusion: This stage spans 120 epochs and focuses on optimizing prompts. The introduction of the modality-aware semantic fusion block integrates textual semantic information, providing high-order semantic supervision signals to enhance feature learning across modalities. Stage 2: Semantic-driven ReID Model Optimization: In this stage, the ReID model is trained for 120 epochs, leveraging the high-order semantic supervision signals generated in Stage 1, along with a dynamic contrastive learning block. Across all stages, the Adam optimizer [54] is used with a cosine learning rate scheduler, starting at $3.5e-4$. The batch size is set to 128. The value of k is set to 5 for the RegDB dataset, 20 for the SYSU-MM01

dataset, and 25 for the LLCM dataset, based on empirical observations. The number of learnable tokens are 4 ($M = 4$). The number of attention heads in MSF is 8. Hyper-parameters are configured as follows: $\lambda = 0.8$, with loss contribution parameters $\alpha_1 = 10.0$ and $\alpha_2 = 1.0$. The detailed analysis can be found in Section IV-F. All methods are implemented using PyTorch on a single NVIDIA A800 80GB GPU.

C. Comparison With State-of-the-Art Methods

In Table I, we evaluate the effectiveness of our proposed Semantic-aware Multimodal Collaborative Learning (SAMCL) framework by comparing it with state-of-the-art supervised and unsupervised VI-ReID methods on two benchmark datasets: SYSU-MM01, RegDB. To further validate the generalization ability under low-light and more challenging modality discrepancy, we additionally report the results on the LLCM dataset in Table II. This comparison highlights the performance gains achieved by our approach in cross-modal person re-identification tasks.

1) *Comparison With Supervised Methods*: We evaluate our approach against ten leading supervised methods, including AGW [5], MSO [19], FMCNet [33], MAUM [23], PartMix [21], DEEN [25] and PMWGCN [34], IDKL [35], EEES [36], CSCL [37]. It can be seen that our SAMCL framework achieves very competitive recognition performance, even outperforms some of these supervised models on both SYSU-MM01 and RegDB datasets. For example, on the large-scale SYSU-MM01 dataset, SAMCL improves Rank-1 accuracy by 2.34% and 1.51% compared to FMCNet [33]. SAMCL achieves competitive results against more advanced models, it comes close to the state-of-the-art PMWGCN [34], with mAP score of 63.05% (vs. 64.90%) and 74.79% (vs. 76.20%) on SYSU-MM01, and 90.85% (vs. 84.50%) and 90.21% (vs. 81.60%) on RegDB. Moreover, it can be seen that the proposed SAMCL outperforms all compared method consistently on the small-scale RegDB dataset. These improvements are primarily driven by our modality-aware semantic fusion module and dynamic contrastive learning that significantly strengthens cross-modal feature alignment and representation, contributing to the overall performance gains.

2) *Comparison With Unsupervised Methods*: As shown in Table I, our Semantic-Aware Multimodal Collaborative Learning framework outperforms a range of state-of-the-art unsupervised VI-ReID methods, including ICE [38], H2H [27], PPLR [39], OTLA [28], ADCA [29], DPIS [40], DOTLA [31], CCLNet [18], EBCMCM [41], PGM [30], GUR [42], SCA-RCP [43], BCGM [44], MMM [45], and RPNR [47]. Specifically, SAMCL achieves a Rank-1 accuracy of 68.68% and an mAP score of 63.05% on SYSU-MM01 (All Search), and 94.47% and 90.85% on RegDB (Visible-to-Infrared). Notably, our method surpasses the state-of-the-art RPNR [47] by 3.48% in Rank-1 accuracy and 3.05% in mAP on SYSU-MM01, as well as by 3.57% in Rank-1 accuracy and 6.15% in mAP on RegDB. While SDCL [46]—which leverages a computationally intensive Transformer-based feature extractor—achieves slightly better performance on certain metrics for the SYSU-MM01 dataset, our approach (using ResNet50 as the feature extractor) remains highly competitive,

TABLE I

THE COMPARISON WITH THE STATE-OF-THE-ART METHODS ON SYSU-MM01 AND REGDB. THE BEST RESULTS ARE PRESENTED IN BOLD FONT. '-' DENOTES THAT NO REPORTED RESULTS ARE AVAILABLE

Type	Methods	Venue	SYSU-MM01						RegDB					
			All Search			Indoor Search			Visible to Infrared			Infrared to Visible		
			r1	mAP	mINP	r1	mAP	mINP	r1	mAP	mINP	r1	mAP	mINP
Supervised	AGW [5]	TPAMI-21	47.50	47.65	35.30	54.17	62.97	59.23	70.05	66.37	50.19	70.49	65.90	51.24
	MSO [19]	MM-21	58.70	56.42	-	63.09	70.31	-	73.60	66.90	-	74.60	67.50	-
	FMCNet [33]	CVPR-22	66.34	62.51	-	68.15	74.09	-	89.12	84.43	-	88.38	83.86	-
	MAUM [23]	CVPR-22	71.68	68.79	-	76.97	81.94	-	87.87	85.09	-	86.95	84.34	-
	PartMix [21]	CVPR-23	77.78	74.62	-	81.52	84.38	-	84.93	82.52	-	85.66	82.27	-
	DEEN [25]	CVPR-23	74.70	71.80	-	80.30	83.30	-	91.10	85.10	-	89.50	83.40	-
	PMWGCN [34]	TIFS-24	66.80	64.90	-	72.60	76.20	-	90.60	84.50	-	88.80	81.60	-
	IDKL [35]	CVPR-24	81.42	79.85	-	87.14	89.37	-	94.72	90.19	-	94.22	90.43	-
	EEES [36]	arXiv-24	78.20	75.70	-	86.50	88.60	-	93.80	88.50	-	94.20	88.20	-
	CSCL [37]	TMM-25	75.72	72.18	-	80.80	83.56	-	92.16	84.27	-	89.66	85.07	-
Unsupervised	ICE [38]	ICCV-21	20.54	20.39	10.24	29.81	38.35	34.32	12.98	15.64	11.91	12.18	14.82	10.60
	H2H [27]	TIP-21	30.15	29.40	-	-	-	-	23.81	18.87	-	-	-	-
	PPLR [39]	CVPR-22	11.98	12.25	4.97	12.71	20.81	17.61	10.30	11.94	8.10	10.39	11.23	7.04
	OTLA [28]	ECCV-22	29.90	27.10	-	29.80	38.80	-	32.90	29.70	-	32.10	28.60	-
	ADCA [29]	MM-22	45.51	42.73	28.29	50.60	59.11	55.17	67.20	64.05	52.67	68.48	63.81	49.62
	DPIS [40]	ICCV-23	58.4	55.6	-	63.00	70.00	-	62.30	53.20	-	61.50	52.70	-
	DOTLA [31]	MM-23	50.36	47.36	32.40	53.47	61.73	57.35	85.63	76.71	61.58	82.91	74.97	58.60
	CCLNet [18]	MM-23	54.03	50.19	-	56.68	65.12	-	69.94	65.53	-	70.17	66.66	-
	EBCMCM [41]	MM-23	53.14	48.16	32.41	55.21	61.98	57.13	83.79	77.87	65.04	82.82	76.74	61.73
	PGM [30]	CVPR-23	57.27	51.78	34.96	56.23	62.74	58.13	69.48	65.41	-	69.85	65.17	-
	GUR [42]	ICCV-23	60.95	56.99	41.85	64.22	69.49	63.81	73.91	70.23	58.88	75.00	69.94	56.21
	SCA-RCP [43]	TKDE-24	51.41	48.52	33.56	56.77	64.19	59.25	85.59	79.12	-	82.41	75.73	-
	BCGM [44]	MM-24	58.90	53.60	36.5	60.30	67.00	62.80	86.10	81.00	67.7	86.50	81.8	70.1
	MMM [45]	ECCV-24	61.60	57.90	-	64.40	70.40	-	89.70	80.50	-	85.80	77.0	-
	SDCL [46]	CVPR-24	64.49	63.24	51.06	71.37	76.90	73.50	86.91	78.92	62.83	85.76	77.25	59.57
	RPNR [47]	MM-24	65.20	60.00	-	68.90	74.40	-	90.90	84.70	-	90.10	83.20	-
Ours	SAMCL	-	68.68	63.05	47.90	69.66	74.79	70.80	94.47	90.85	84.15	94.13	90.21	82.05

Note: Results of SDCL are obtained using a Transformer-based backbone; ours (SAMCL) are based on ResNet-50.

TABLE II

THE COMPARISON WITH THE STATE-OF-THE-ART METHODS ON LLCM

Type	Methods	Venue	Visible to Infrared		Infrared to Visible	
			r1	mAP	r1	mAP
Supervised	CA [9]	ICCV-21	56.5	59.8	48.8	56.6
	AGW [5]	TPAMI-21	51.5	55.3	43.6	51.8
	DART [48]	CVPR-22	60.4	63.2	52.2	59.8
	DEEN [25]	CVPR-23	62.5	65.8	54.9	62.9
	HOS-Net [49]	AAAI-24	64.9	67.9	56.4	63.2
Unsupervised	OTLA [28]	ICCV-21	17.8	20.4	14.9	18.6
	ADCA [29]	MM-22	23.5	28.2	16.1	21.4
	CCLNet [18]	MM-23	45.3	49.9	39.3	45.3
	PGM [30]	CVPR-23	44.4	48.6	38.4	44.2
	SDCL [46]	CVPR-24	46.9	52.4	43.4	48.2
	SCA-RCP [43]	TKDE-24	29.1	33.3	22.3	28.0
	Ours	-	50.6	54.3	45.1	49.3

and it outperforms SDCL on the RegDB dataset. These results highlight SAMCL's consistent superiority over other unsupervised methods, confirming its effectiveness across different datasets. As further evidenced by Table II, SAMCL achieves state-of-the-art unsupervised performance on the challenging low-light LLCM benchmark. In the Visible-to-Infrared setting, SAMCL reaches 50.6% Rank-1 and 54.3% mAP, outperforming SDCL (46.9%/52.4%) by + 3.7% in Rank-1 and + 1.9% in mAP. In the more challenging Infrared-to-Visible setting, SAMCL achieves 45.1% Rank-1 and 49.3% mAP, surpassing SDCL (43.4%/48.2%) by + 1.7% Rank-1 and + 1.1% mAP. These consistent gains across both retrieval directions demonstrate superior robustness of SAMCL under low-light conditions, where illumination degradation further amplifies the cross-modality discrepancy. Overall, the consistent gains

on SYSU-MM01, RegDB, and LLCM validate the robustness and transferability of SAMCL across diverse cross-modality conditions. One key advantage of SAMCL lies in its use of semantic supervision, which provides more reliable optimization signals compared to rigid pseudo-labels. By reducing the dependence on noisy clustering pseudo-labels, SAMCL offers a more effective and stable solution for unsupervised VI-ReID.

D. Ablation Study

In this subsection, we conduct ablation experiments to evaluate the effectiveness of our proposed method. (1) Effectiveness of each component: We analyze the contributions of L_{DCL} and L_{prompt} (Eq. 16) to recognition performance on the SYSU-MM01 and RegDB datasets. Using a baseline unsupervised VI-ReID model optimized with L_{id} and L_{tri} , we isolate and quantify the individual impact of these components. (2) Effectiveness of modality-aware semantic fusion (MSF): We evaluate the MSF module's role in enhancing performance during the prompt generation and ReID model optimization stages. The results, visualized in Fig. 3, highlight significant performance gains achieved by integrating MSF. (3) Effectiveness of dynamic contrastive learning (DCL): We further validate the effectiveness of the DCL module, which is designed to reduce intra-modality variations and enhance the robustness of clustering pseudo-labels. As shown in Fig. 4, its impact is assessed by analyzing the reliability of clustering pseudo-labels.

1) *Effectiveness of Each Component*: In this subsection, we validate the effectiveness of the dynamic contrastive learning loss (L_{DCL}) and the semantic-guided loss (L_{prompt}) introduced

TABLE III

ABLATION STUDY: EFFECTIVENESS OF EACH COMPONENT ON THE SYSU-MM01 AND REGDB DATASETS. L_{DCL} IS DERIVED FROM EQ. (12) AND L_{prompt} DERIVED FROM EQ. (15). RANK-1 ACCURACY(%), MAP (%) AND MINP (%) ARE REPORTED

Index	Module			SYSU-MM01(All)			SYSU-MM01(Indoor)			RegDB(V2I)			RegDB(I2V)		
	Baseline	L_{DCL}	L_{prompt}	r1	mAP	mINP	r1	mAP	mINP	r1	mAP	mINP	r1	mAP	mINP
1	✓			48.47	44.38	29.33	50.11	45.23	30.23	34.51	33.13	25.76	33.98	33.66	24.90
2		✓		50.23	47.34	31.44	53.12	47.78	34.77	63.69	58.20	46.29	63.98	57.66	45.28
3	✓		✓	61.56	55.50	38.12	64.56	71.15	67.61	80.99	76.95	67.87	81.23	77.87	67.90
4	✓	✓	✓	68.68	63.05	47.90	69.66	74.79	70.80	94.47	90.85	84.15	94.13	90.21	82.05

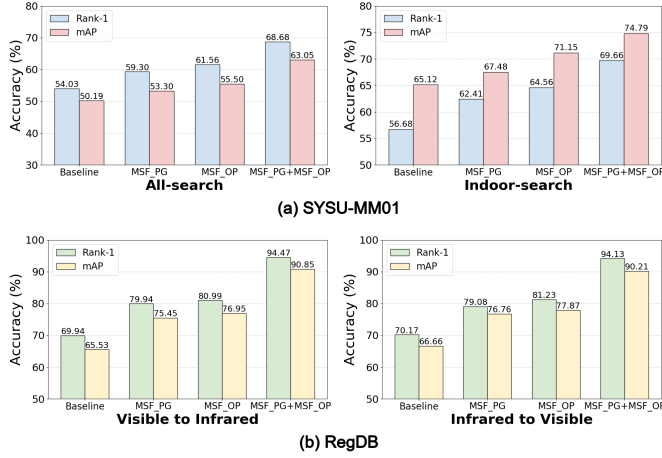


Fig. 3. An ablation study of the MSF module during the prompt generation stage (MSF_PG) and the semantic-driven ReID model optimization stage (MSF_OP). (a) shows the experimental results on the SYSU-MM01 dataset, while (b) presents the results on the RegDB dataset.

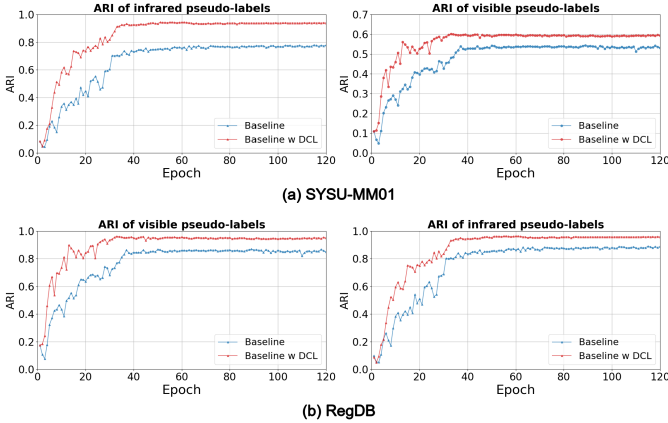


Fig. 4. Illustration of the effectiveness of the DCL module in enhancing pseudo-label quality on (a) SYSU-MM01 and (b) RegDB. The Adjusted Rand Index (ARI) quantifies the clustering accuracy of pseudo-labels between visible and infrared modalities (a higher ARI indicates better alignment of clustering results with true labels).

in Eq. (16) on the SYSU-MM01 and RegDB datasets. As shown in Table III, integrating L_{DCL} improves Rank-1 accuracy on the challenging SYSU-MM01 dataset by 1.76% in the all-search mode and 3.01% in the indoor-search mode. This improvement demonstrates L_{DCL} 's ability to reduce the impact of clustering noise during training and enhance robustness against intra-modal variations, significantly boosting recognition performance in unsupervised visible-infrared person re-identification tasks.

On the other hand, L_{prompt} , which incorporates textual class semantic information as supervision signals, reduces dependence on noisy clustering pseudo-labels. This results in a substantial improvement, with Rank-1 accuracy increasing by 13.09% in all-search mode and 14.45% in indoor-search mode. By leveraging rich semantic information, L_{prompt} enhances the alignment between cross-modal features and facilitates more reliable supervision.

When L_{DCL} and L_{prompt} are combined, the model achieves its best performance, with Rank-1 accuracy increasing by 20.21% in all-search mode and 19.55% in indoor-search mode. These results highlight the complementary nature of the two components: L_{DCL} focuses on improving robustness during feature learning, while L_{prompt} strengthens semantic alignment, making them jointly effective for unsupervised visible-infrared person re-identification.

2) *Effectiveness of MSF Module*: The modality-aware semantic fusion (MSF) module integrates complementary textual features from visible and infrared modalities to strengthen cross-modal supervision, delivering robust signals for collaborative training and significantly enhancing recognition performance. To evaluate its effectiveness, we test the MSF module during the prompt generation (MSF_PG) in Stage 1 and the ReID model optimization (MSF_OP) in Stage 2. Fig. 3 shows the performance gains on SYSU-MM01 and RegDB. Here, the “Baseline” refers to our method without MSF.

Integrating MSF during the prompt generation stage (MSF_PG) improves Rank-1 accuracy on the SYSU-MM01 dataset by 5.27% and mAP by 8.92% for all-search mode, with corresponding improvements of 5.73% and 2.36% for indoor-search mode. Incorporating MSF in the ReID model optimization stage (MSF_OP) achieves even higher gains, with Rank-1 accuracy increasing by 7.53% and mAP by 11.21% for all-search mode, and by 7.88% and 6.30% for indoor-search.

Combining MSF_PG and MSF_OP delivers the best results, improving Rank-1 accuracy by 9.38% over MSF_PG alone and by 7.12% over MSF_OP for all-search mode on SYSU-MM01. Similar trends are observed on the RegDB dataset, demonstrating the consistent effectiveness of the MSF module. The MSF module enhances the representation and generalization of textual semantic information, generating robust semantic features for model optimization. This significantly boosts the model's recognition performance in VI-ReID tasks.

3) *Effectiveness of Dynamic Contrastive Learning*: To assess the effectiveness of the proposed DCL module, we conduct a detailed pseudo-label evaluation experiment on SYSU-MM01 and RegDB. The Adjusted Rand Index (ARI)

TABLE IV

COMPARISONS WITH OTHER CONTRASTIVE LEARNING METHODS. “BASELINE” DENOTES OUR METHOD WITHOUT CONTRASTIVE LEARNING

Methods	Rank-1	mAP
Baseline	61.56	55.50
Ours with InfoNCE loss [55], [56]	66.43	62.78
Ours with Contrastive loss [57]	65.89	61.44
Ours with DCL	68.68	63.05

[58] is employed as the evaluation metric to quantify the accuracy of pseudo-labels generated for both visible and infrared modalities. ARI measures the similarity between the clustering results and the ground truth labels, where higher scores indicate more accurate intra-modal clustering and better pseudo-label quality.

In this experiment, the baseline model is optimized with $L_{id} + L_{tri} + L_{prompt}$ loss functions, while the DCL module is added as an enhancement to mitigate intra-modal variations and reduce noise in the clustering process. As shown in Fig. 4, the introduction of the DCL module leads to significant improvements in ARI scores: a 6% increase for the visible modality and a notable 16% increase for the infrared modality on the SYSU-MM01 dataset, while both visible and infrared modalities demonstrate consistent 6% improvements on the RegDB dataset with the ARI value reaching 95.7%, which is close to the ground truth. This demonstrates that DCL effectively refines the clustering process by reducing label noise and improving the alignment of intra-modal features. The enhanced clustering precision facilitated by DCL contributes to more robust pseudo-labels, which in turn improve the model’s overall stability and recognition performance.

To further validate the DCL module’s effectiveness, we replace it within SAMCL with static pair-based contrastive methods, e.g., MoCo [55], SimCLR [56], DrLIM [57]. As Table IV shows (SYSU-MM01, All-Search), DCL’s dynamic mechanism significantly enhances feature representation capability and outperforms all compared methods consistently. Specifically, DCL achieves 68.68% Rank-1/63.05% mAP—outperforming InfoNCE (66.43%/62.78%) and Contrastive loss (65.89%/61.44%). This stems from: (1) Dynamic k-NN embeddings adapting to evolving feature distributions; (2) Similarity-based weighting suppressing noisy clusters; (3) Margin-based repulsion correcting pseudo-label errors. These innovations address intra-modality variance through dynamic adaptation and noise-aware optimization, driving SAMCL’s SOTA cross-modality performance.

E. Prompt Analysis

This subsection demonstrates the critical role of prompt design in unsupervised cross-modal person re-identification through three dimensions: prompt template architecture, learnable token quantity, and semantic visualization.

1) *Prompt Template Architecture*: Our fixed prompt structure (“A [modality] photo of a $[X_i]_1[X_i]_2 \dots [X_i]_M$ person”) leverages CLIP’s cross-modal architecture to address VI-ReID requirements, where explicit modality anchors (“visible”/“infrared”) serve dual functions: (1) activating CLIP’s modality-discriminative capabilities; (2) providing

TABLE V

PERFORMANCE OF OUR METHOD WITH DIFFERENT PROMPT TEMPLATES

Methods	Rank-1	mAP
Baseline	50.23	47.34
$[X_i]_1[X_i]_2 \dots [X_i]_M$	63.43	58.78
A photo of a $[X_i]_1[X_i]_2 \dots [X_i]_M$ person	66.22	60.09
A visible/infrared photo of a $[X_i]_1[X_i]_2 \dots [X_i]_M$ person	68.68	63.05

TABLE VI

RECOGNITION PERFORMANCE OF OUR METHOD WITH VARIOUS NUMBER OF LEARNABLE TOKENS

M	Rank-1	mAP
1	66.01	60.12
2	67.34	61.88
4	68.68	63.05
8	67.92	62.17
16	65.85	60.03

semantic framework that focuses learnable tokens $[X_i]$ exclusively on identity feature optimization, eliminating ambiguity. This enhances modality-specific representation learning for improved cross-modality matching, as validated on SYSU-MM01 (all-search, Table V, “Baseline” denotes our method without MSF): Modality-agnostic prompts (“A photo of a $[X_i]_1[X_i]_2 \dots [X_i]_M$ person”) reduce Rank-1/mAP by 2.46%/2.96% due to cross-modal confusion, while globally learnable prompts (“ $[X_i]_1 \dots [X_i]_M$ ”) degrade performance by 5.25%/4.27% from oversimplified identity descriptions.

2) *The Number of the Learnable Tokens*: To validate the effectiveness of our prompts, in this subsection, we test the performance of our method with different numbers of learnable tokens, i.e., M , within the prompt template (e.g., “A visible/infrared photo of a $[X_i]_1[X_i]_2 \dots [X_i]_M$ person”). These tokens generate identity-related semantic features through CLIP’s text encoder, which are then utilized by the Modality-aware Semantic Fusion module to produce cross-modality supervision signals. To evaluate the impact of M on model performance, as shown in Table VI, we conduct an ablation study on SYSU-MM01 under the all-search setting to reveal critical performance trade-offs: Oversimplified semantic descriptions at $M=1$ degrade cross-modality matching due to underfitting of identity features (Rank-1/mAP: 62.34%/56.78%). Conversely, excessive tokens ($M \geq 8$) introduce redundant complexity that distracts optimization and increases overfitting risk ($M=8$: 66.12%/60.45%). Optimal performance occurs at $M=4$ (Rank-1/mAP: 68.68%/63.05%), where token complexity balances expressive identity representation with training stability.

3) *Pseudo-to-Real Textual Description Retrieval*: In this subsection, we perform a qualitative and quantitative analysis to thoroughly evaluate the semantic capture capability of the learnable tokens using a curated pedestrian attribute vocabulary. It is organized into four categories: Actions (Run, Jump, Write, Read, Walk), Colors (Red, Black, Green, Yellow, Purple, Pink, Gray, Black, White), Age and Gender (Child, Teenager, Adult, Male, Female), and Clothing and Accessories (Bag, Backpack, Hat, Glasses, Watch, Tie, Shoes, Shirt, Pants, Dress, Skirt, Jacket, T-shirt, Jeans, Shorts). For each token $[X_i]$ (representing the i -th cluster/identity), we compute its cosine

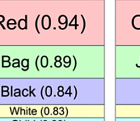
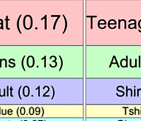
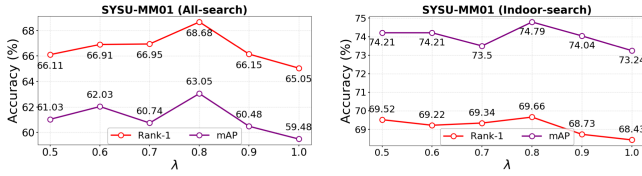
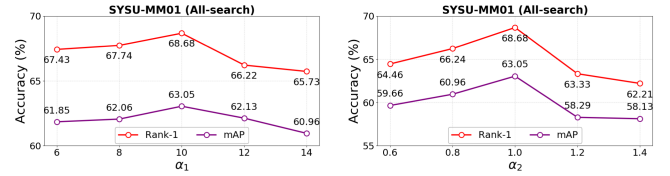
Identification	$[X_1]$		$[X_2]$		$[X_3]$		$[X_4]$	
	Epoch 0	MSF	Epoch 0	MSF	Epoch 0	MSF	Epoch 0	MSF
								
ID→#305	Run (0.19)	Red (0.94)	Coat (0.17)	Teenager (0.93)	Yellow (0.23)	Black (0.95)	Gloves (0.18)	Male (0.93)
	Green (0.15)	Bag (0.89)	Jeans (0.13)	Adult (0.86)	Male (0.14)	Red (0.91)	Skirt (0.15)	White (0.84)
ID→#305	Shirt (0.11)	Black (0.84)	Adult (0.12)	Shirt (0.83)	Dress (0.11)	Male (0.88)	Sing (0.13)	Shoes (0.81)
	Dance (0.10)	White (0.83)	Blue (0.09)	Tshirt (0.82)	Scarf (0.09)	Jeans (0.82)	Orange (0.08)	Tshirt (0.78)
ID→#305	Hat (0.06)	Child (0.80)	Pants (0.07)	Glasses (0.79)	Jump (0.05)	White (0.77)	Female (0.07)	Jeans (0.77)
ID→#305	Socks (0.23)	Walk (0.94)	Pink (0.17)	Tshirt (0.95)	Eat (0.19)	Glasses (0.92)	Sleep (0.17)	Teenager (0.93)
	Write (0.16)	Run (0.88)	Teenager (0.15)	Shirt (0.90)	Brown (0.14)	Pants (0.87)	Male (0.15)	Backpack (0.88)
ID→#305	Jacket (0.13)	Bag (0.85)	Watch (0.12)	Walk (0.84)	Adult (0.11)	Bag (0.83)	Tie (0.13)	Male (0.84)
	Child (0.10)	Tshirt (0.81)	Read (0.07)	Backpack (0.82)	Ring (0.09)	Shoes (0.80)	Tshirt (0.08)	Pants (0.81)
ID→#305	Purple (0.05)	Backpack (0.79)	Coat (0.06)	Watch (0.77)	Sweater (0.09)	Walk (0.78)	Gray (0.07)	Jacket (0.80)

Fig. 5. Pseudo-to-real textual description retrieval of the learnable tokens.

Fig. 6. Evaluation of the hyperparameter λ in Eq. (3) and Eq. (4). The results are based on the SYSU-MM01 dataset, with the left figure showing all-search performance and the right figure showing indoor-search performance.Fig. 7. Evaluation of the hyperparameter α_1 and α_2 in Eq. (16) on the SYSU-MM01 dataset in all-search mode.

similarity with every vocabulary word. The top-5 matching words, along with their relevance scores, are extracted as semantic descriptors. Figure 5 visualizes these semantic descriptions for tokens corresponding to visible (VIS) and infrared (IR) modalities of IDs 305 and 217 from the SYSU-MM01 dataset. Word clouds depict the descriptors at two training stages: initial state (Epoch 0) and after optimization with the MSF module (Epoch 120).

Quantitatively, the semantic relevance undergoes a significant evolution: initial descriptors show low relevance (Epoch 0 scores < 0.25 , e.g., “Run” = 0.19 for ID305-VIS, inconsistent with content), while training yields precise semantic alignment (Epoch 120 scores > 0.8 , e.g., “Bag” = 0.93 and “Walk” = 0.89 for ID217-IR). Qualitatively, Figure 5 reveals two key observations: (1) Modality-specific semantics: VIS tokens emphasize color attributes (e.g., “Red”/“Black” for ID305-VIS), while IR tokens focus on structural elements (e.g., “Glasses”/“Bag” for ID305-IR). (2) Cross-modal identity consistency: Complementary attributes across modalities define identity (e.g., “Red” (VIS) and “Glasses” (IR) for ID305). Therefore, the MSF module successfully transforms initially noisy token semantics into semantically rich descriptors (> 0.8 relevance), organically capturing distinct modality features while preserving coherent identity representations across modalities.

F. Hyperparameter Analysis

In this subsection, we analyze the effect caused by hyperparameters, such as λ in Eqs. (3) and (4) for modality-aware prompt fusion, and α_1 and α_2 in Eq. (16) for semantic-driven ReID model optimization.

1) λ in Modality-Aware Semantic Fusion: As reported in Fig. 6, the experimental results on the SYSU-MM01 dataset,

under both All-search and Indoor-search scenarios, highlight the significant impact of the lambda (λ) parameter on model performance. In the All-search scenario, performance peaks at $\lambda = 0.8$, with a Rank-1 accuracy of 68.68% and an mAP of 63.05%, followed by a decline as λ increases. Similarly, in the Indoor-search condition, the highest Rank-1 accuracy of 69.66% and mAP of 74.79% are also achieved at $\lambda = 0.8$. These results reveal a consistent trend: performance improves initially with increasing λ , reaches an optimum at $\lambda = 0.8$, and then declines. The model performs better in the Indoor-search scenario, likely due to reduced environmental interference.

2) α_1 and α_2 in Semantic-Driven ReID Model Optimization: The performance analysis on the SYSU-MM01 dataset under the All-search mode highlights the critical impact of parameters α_1 and α_2 on the model’s Rank-1 accuracy and mAP. As shown in Fig. 7, for α_1 , the optimal value is 10, where Rank-1 accuracy peaks at 68.88% and mAP at 63.05%, showing a gradual increase before declining slightly. Similarly, for α_2 , the best performance occurs at $\alpha_2 = 1.0$, achieving the highest Rank-1 accuracy of 68.68% and mAP of 63.05%, with a sharp decline afterward. The findings emphasize the importance of fine-tuning α_1 and α_2 to achieve optimal recognition results, with $\alpha_2 = 1.0$ and $\alpha_1 = 10$ being the most effective settings for this dataset.

G. Visualization Analysis

To further demonstrate the effectiveness of the proposed MSF and DCL modules, we conduct an extensive t-SNE-based cluster analysis (Fig. 8) comparing four configurations: Baseline (only) exhibits significant modality dispersion where visible (circles) and infrared (triangles) clusters of the same identity (e.g., pink cluster) scatter widely; Baseline + DCL improves intra-modality compactness (tighter clusters) but

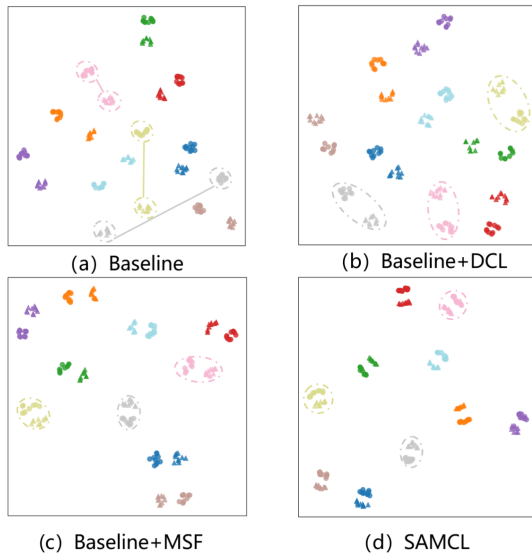


Fig. 8. The t-SNE visualization of 10 randomly selected identities, where each identity is represented by a unique color. Circles denote identities in the visible modality, while triangles represent identities in the infrared modality.

fails to align cross-modality clusters (e.g., green cluster remains split); Baseline + MSF dramatically reduces inter-modality distance causing visible/infrared features of identical identities (e.g., purple cluster) to converge, validating textual supervision's alignment role; Full SAMCL (MSF+DCL) achieves optimal simultaneous intra-modality compactness and cross-modality alignment (overlapping circles/triangles in red cluster). Key insights reveal MSF functions as a semantic bridge: by fusing unimodal prompts into unified descriptors via cross-attention, it creates shared semantic anchors that force visual features toward modality-invariant regions, closing cross-modal gaps (Fig. 8c vs. 8a). The DCL-MSF synergy is critical—DCL refines intra-modality cohesion (Fig. 8d vs. 8c) while MSF ensures cross-modality consistency.

V. CONCLUSION AND FUTURE WORK

Conclusion: In this paper, we introduce a Semantic-Aware Multimodal Collaborative Learning (SAMCL) framework for unsupervised visible-infrared person re-identification (VI-ReID), aiming to minimize dependence on costly cross-modal annotations. To bridge cross-modal discrepancies, we propose a Modality-aware Semantic Fusion (MSF) module, which incorporates single-modality textual semantics to generate enriched high-level semantic supervisory signals. Furthermore, we design a Dynamic Contrastive Learning (DCL) module that dynamically refines feature space relationships by adjusting sample distances relative to their neighbors, thereby enhancing robustness against noisy clustering pseudo-labels. Comprehensive experiments on the SYSU-MM01, RegDB, and LLCM datasets demonstrate that our method not only outperforms existing approaches but also achieves performance comparable to certain supervised methods, underscoring its effectiveness for practical VI-ReID scenarios.

Future Work: Building on the current framework, we outline several promising directions for advancing unsupervised VI-ReID. First, we will explore adaptive neighborhood

mechanisms in Dynamic Contrastive Learning to dynamically adjust the k-nearest neighbors based on sample density, further improving pseudo-label robustness. Second, we aim to generalize the Modality-aware Semantic Fusion module to other cross-modal tasks (e.g., text-image retrieval) while optimizing its computational efficiency. Third, we will extend the framework to extreme scenarios (e.g., low-light or occluded environments) by integrating multi-sensor data (e.g., infrared/mmWave) for more robust matching. These efforts will focus on reducing annotation dependency, enhancing modality invariance, and bridging the gap between research and practical applications.

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